



ECE/CS/ME 539
Introduction to Artificial Neural Networks

Advanced Topics: Object Detection and Segmentation



Computer Vision Tasks

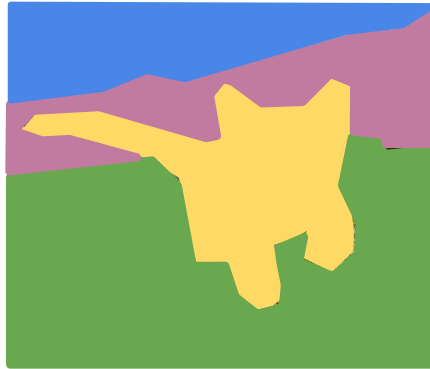
Classification



CAT

No spatial extent

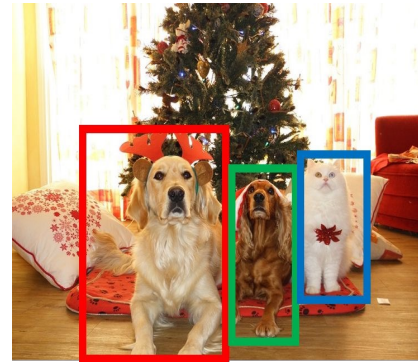
Semantic Segmentation



GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation



DOG, DOG, CAT

[This image is CC0 public domain](#)



Computer Vision Tasks

Classification



CAT

No spatial extent

Semantic Segmentation

Goal: Classify the most prominent object in the image.

We've seen many architectures for classification:

- CNNs (AlexNet, VGG, ResNet, DenseNet, etc)
- Vision Transformers

GRASS, CAT,
TREE, SKY

No objects, just pixels

Object Detection

DOG, DOG, CAT

Multiple Object

Instance Segmentation

DOG, DOG, CAT

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Computer Vision Tasks

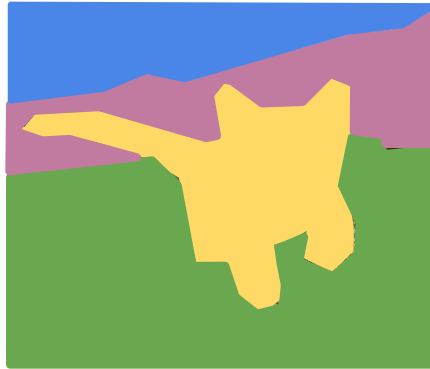
Classification



CAT

No spatial extent

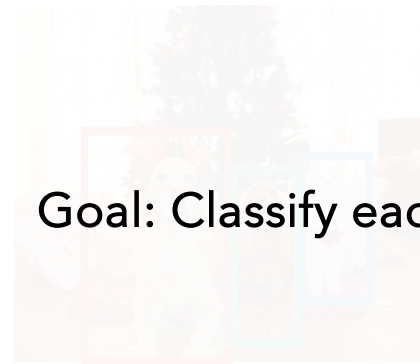
Semantic Segmentation



GRASS, CAT,
TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation



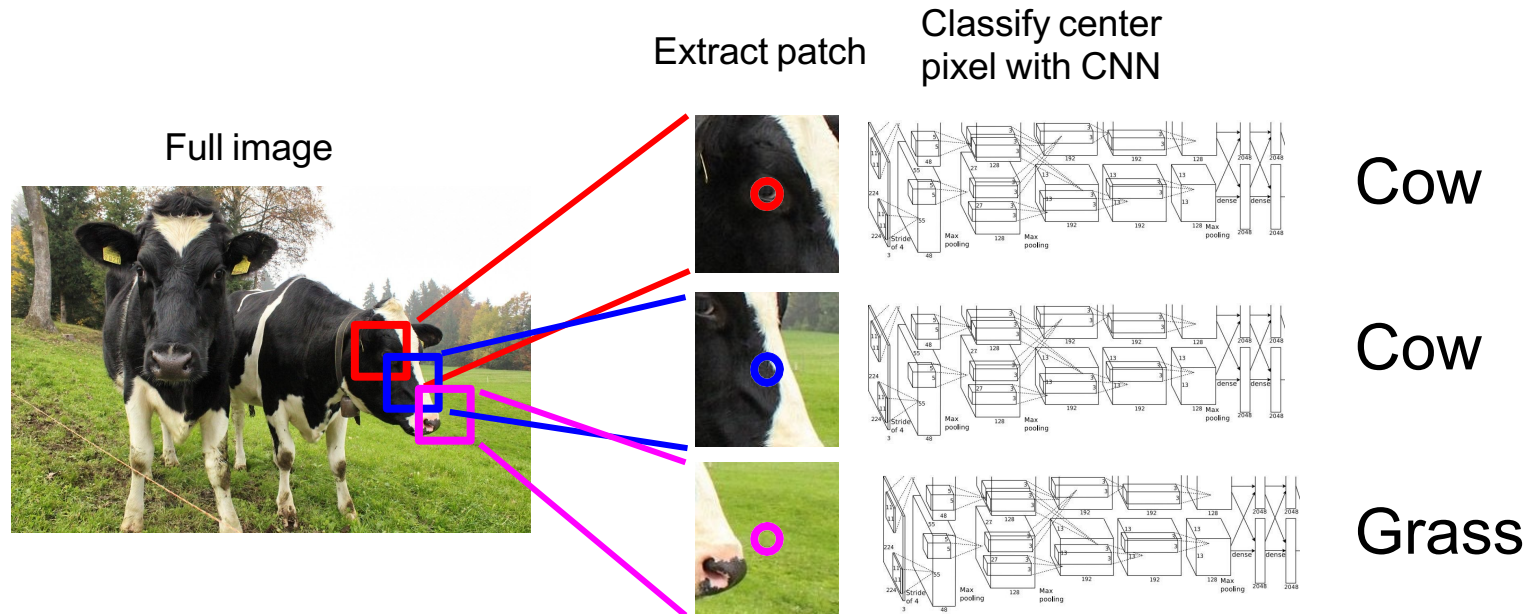
DOG, DOG, CAT

Goal: Classify each pixel of an image.

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Semantic Segmentation Idea: Sliding Window



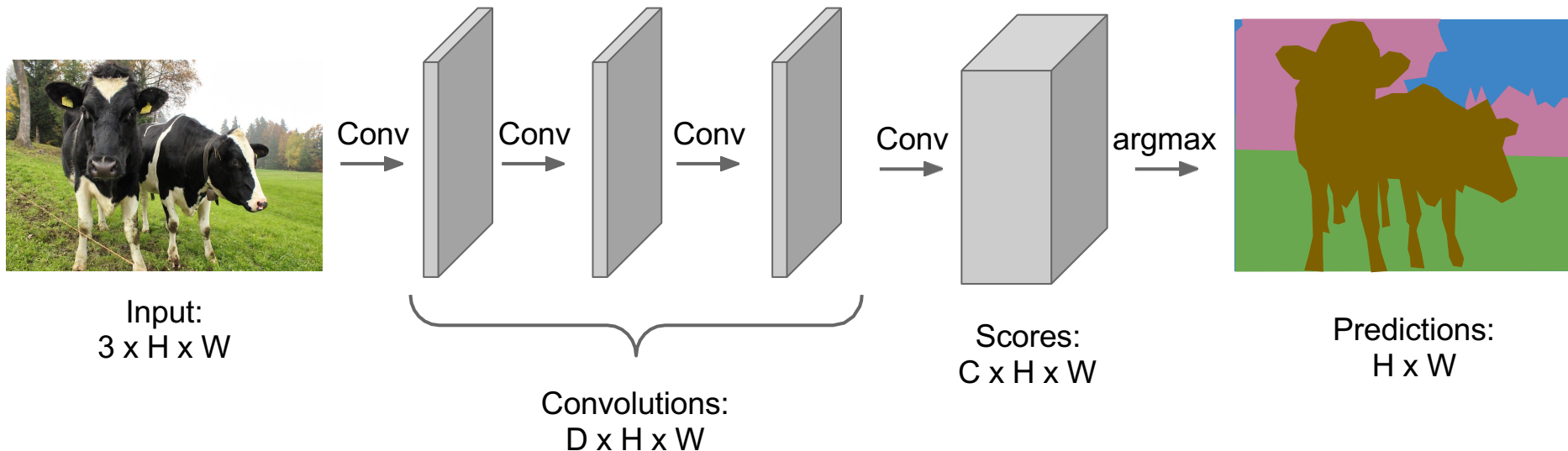
Problem: Very inefficient!

Farabet et al, "Learning Hierarchical Features for Scene Labeling," TPAMI 2013
Pinheiro and Collobert, "Recurrent Convolutional Neural Networks for Scene Labeling", ICML 2014



Semantic Segmentation Idea: Fully Convolutional

Idea: Design a network with only convolutional layers without downsampling operators to make predictions for pixels all at once!



Problem: Convolutions in high resolution are very computationally expensive!

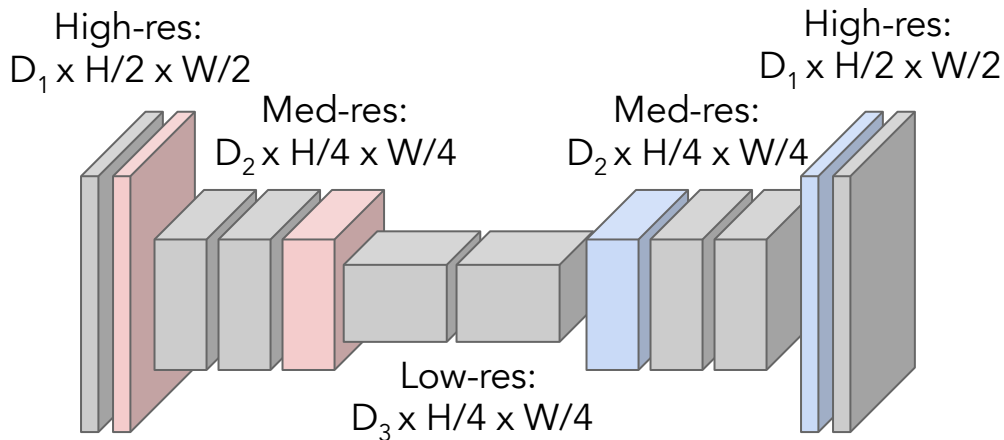


Semantic Segmentation Idea: Fully Convolutional

Idea: Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!



Input:
 $3 \times H \times W$



Downsampling:
Pooling, strided
convolution

Upsampling:
Unpooling or strided
transposed convolution



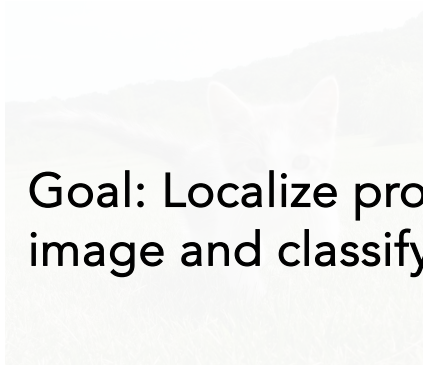
Predictions:
 $H \times W$



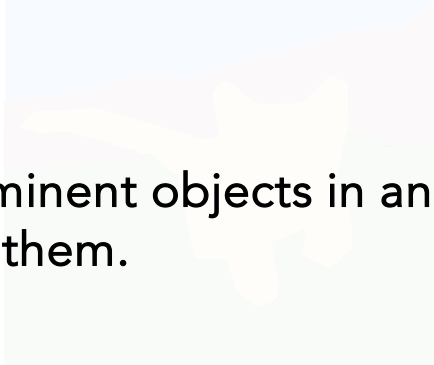
Computer Vision Tasks

Object Detection

Goal: Localize prominent objects in an image and classify them.



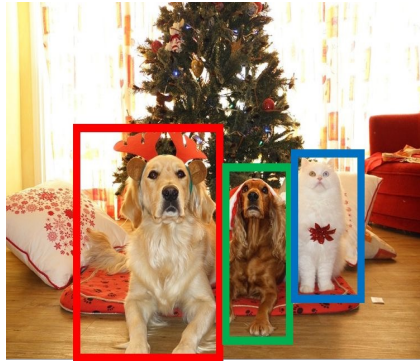
Semantic Segmentation



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No objects, just pixels

Object Detection



DOG, DOG, CAT

Instance Segmentation



DOG, DOG, CAT

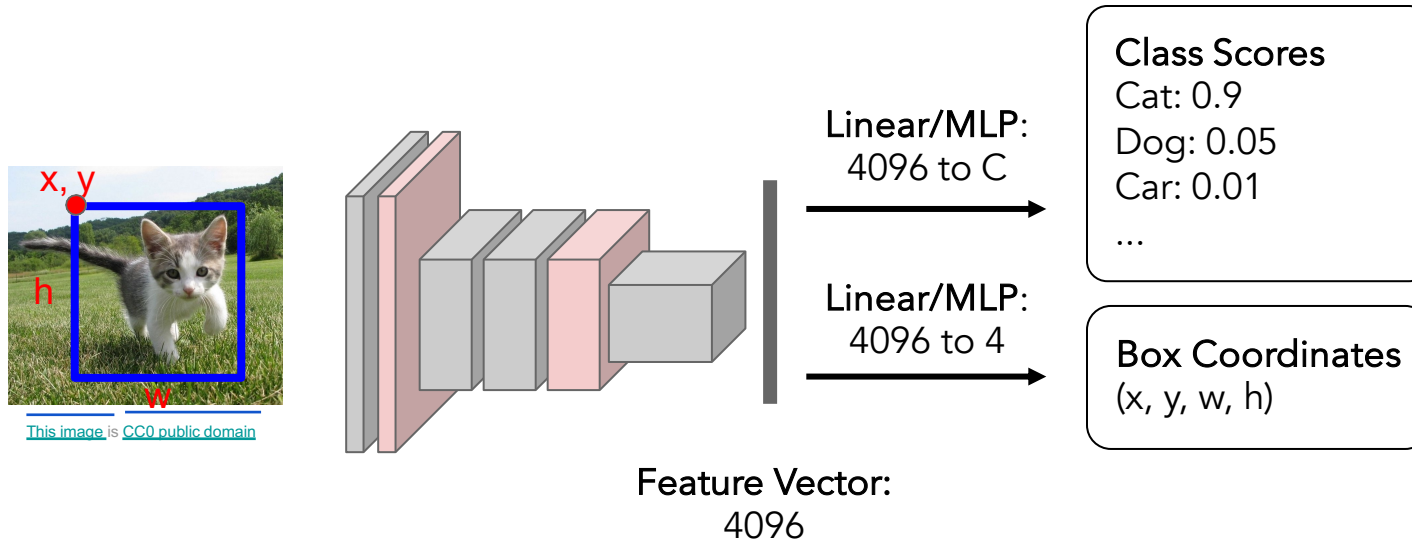
Multiple Object

[This image is CC0 public domain](#)



Object Detection

(Classification + Localization)



Treat localization as a regression problem
(MSE Loss):

$$L_{loc} = \left\| \begin{bmatrix} x \\ y \\ w \\ h \end{bmatrix} - \begin{bmatrix} x^* \\ y^* \\ w^* \\ h^* \end{bmatrix} \right\|^2$$

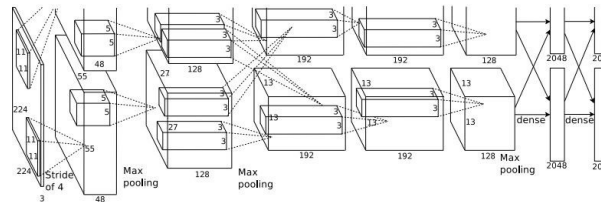
Classification learned by cross-entropy:

$$L_{cls} = -\log P(class^* | crop)$$

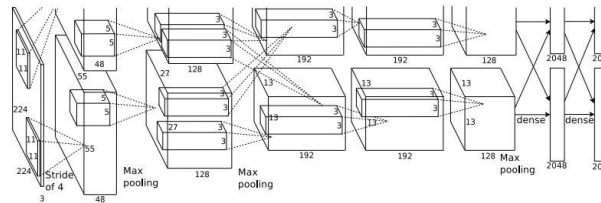


Object Detection: Multiple Objects

Problem: Each image needs a different number of predictions.



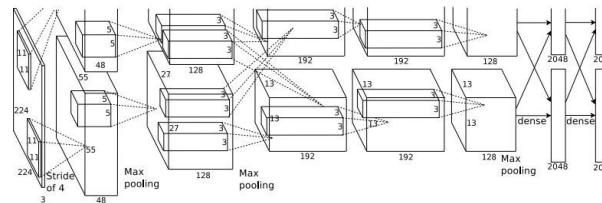
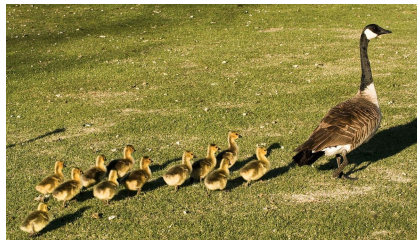
CAT: (x, y, w, h)



DOG: (x, y, w, h)

DOG: (x, y, w, h)

CAT: (x, y, w, h)



DUCK: (x, y, w, h)

DUCK: (x, y, w, h)

DUCK: (x, y, w, h)

...



Object Detection: Multiple Objects

Idea: Apply a model separately to many crops. For each crop classify as an object or background.



Dog? NO
Cat? NO
Background? YES



Dog? YES
Cat? NO
Background? NO



Dog? NO
Cat? YES
Background? NO



R-CNN

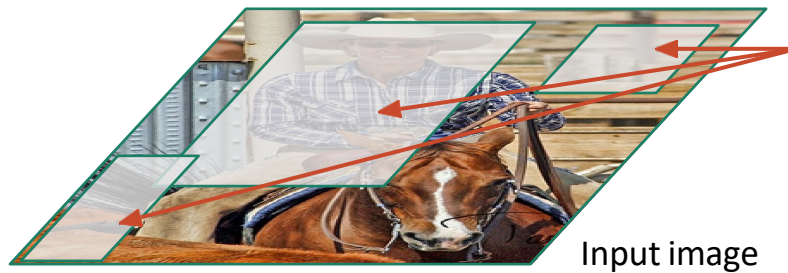


Input image

Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014. [source](#). Reproduced with permission.



R-CNN

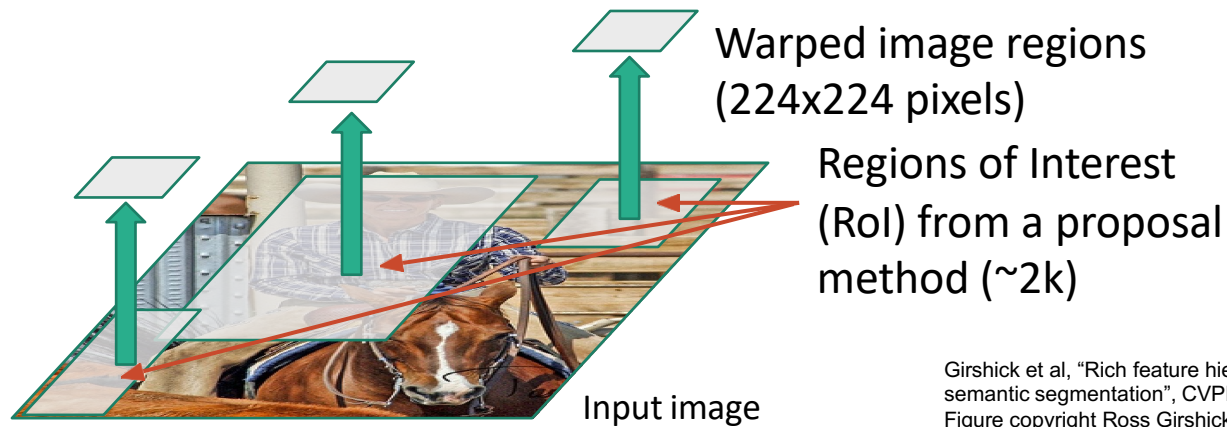


Regions of Interest
(RoI) from a proposal
method (~2k)

Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014. [source](#). Reproduced with permission.



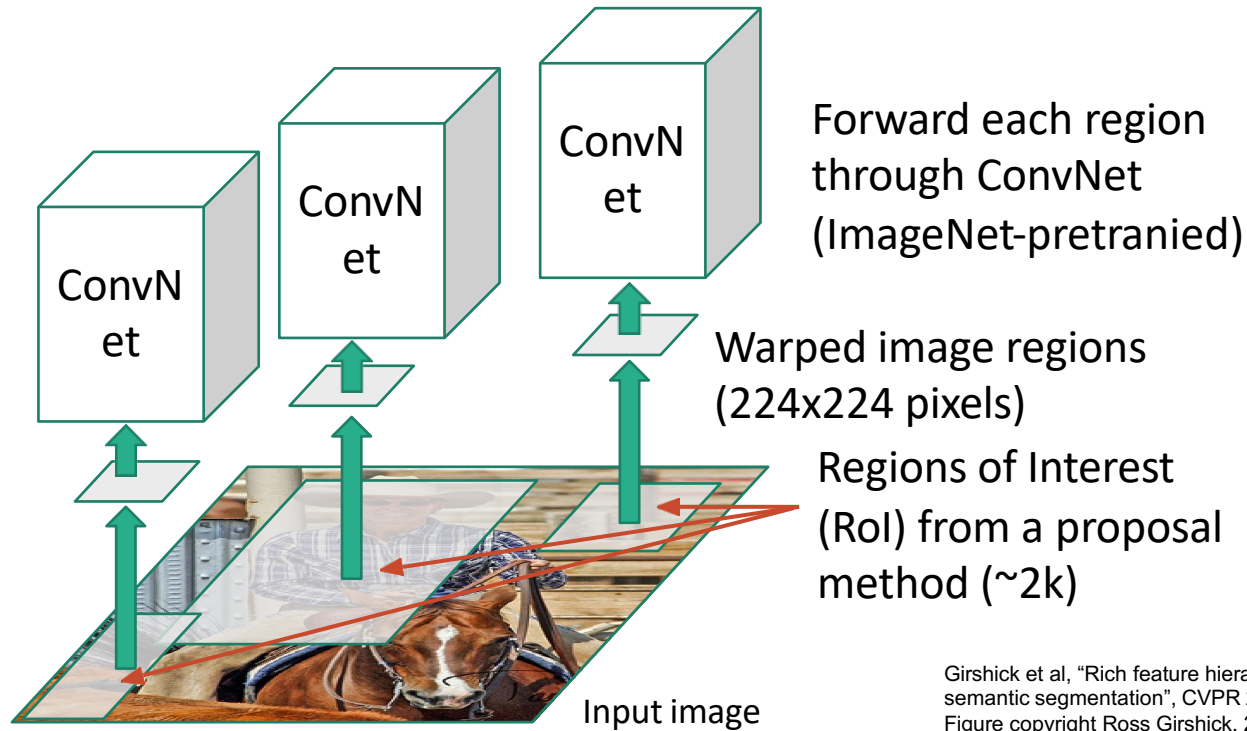
R-CNN



Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014. [source](#). Reproduced with permission.



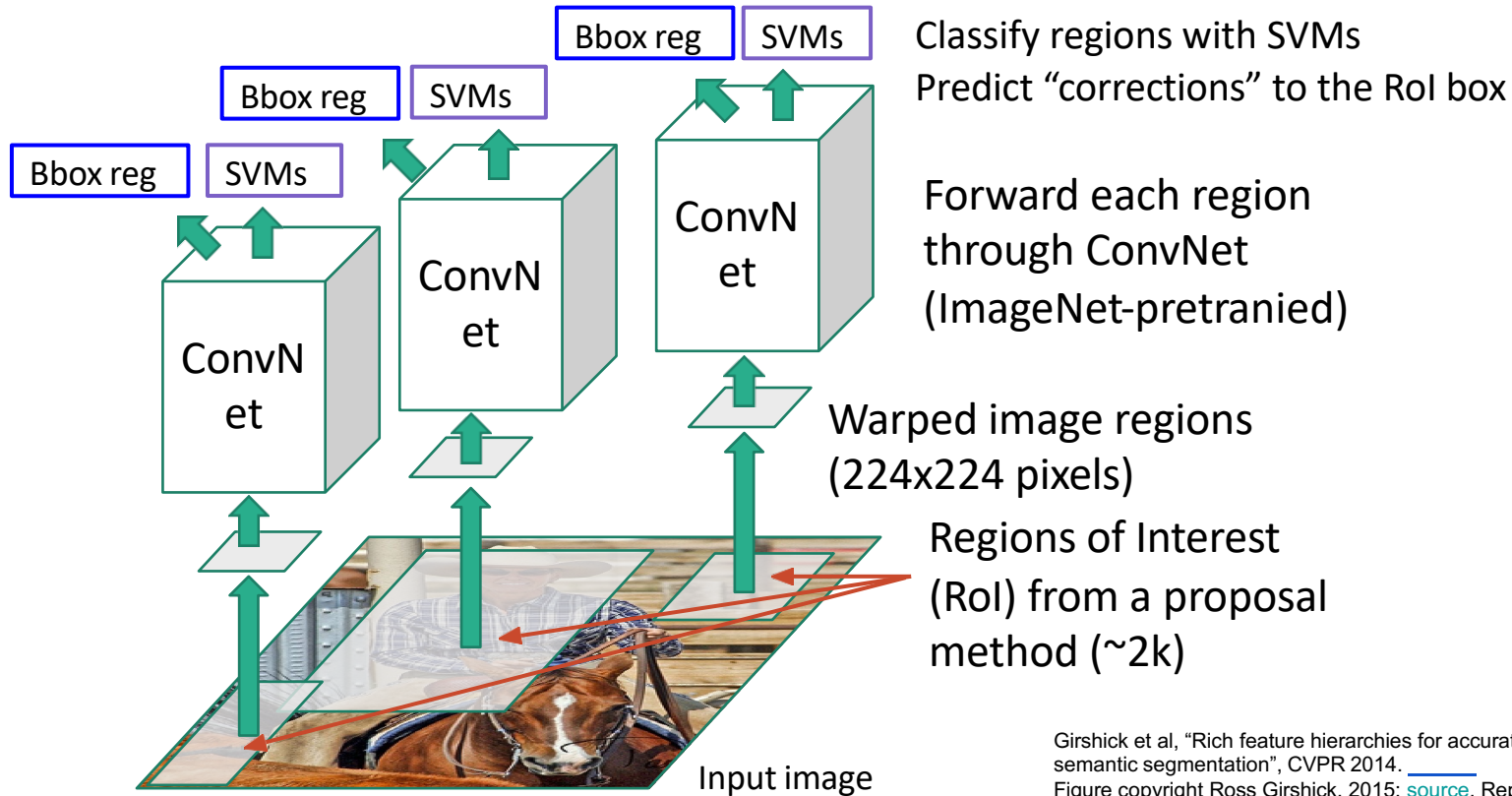
R-CNN



Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014. [source](#). Reproduced with permission.



R-CNN

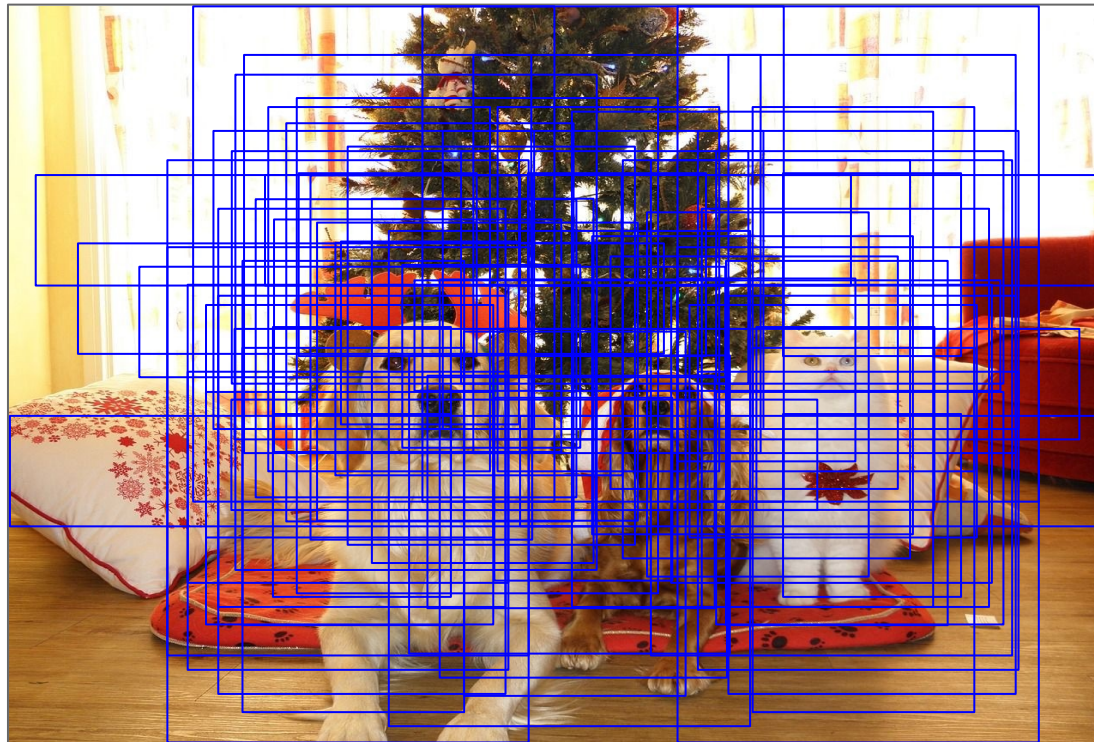


Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014. [source](#). Reproduced with permission.



R-CNN

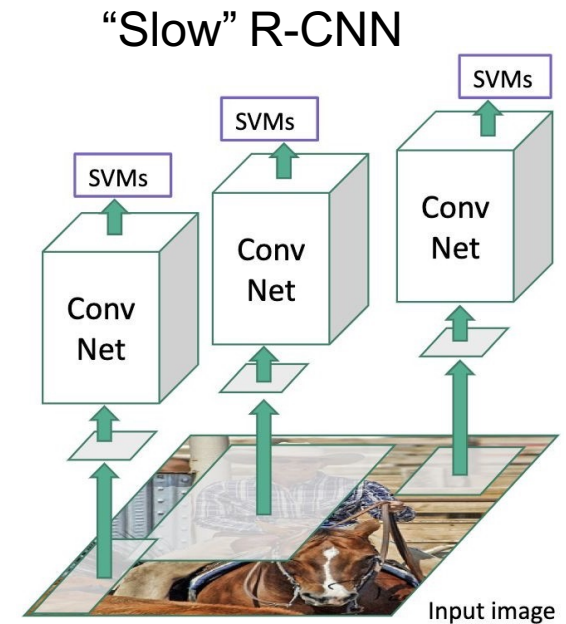
Problem: There is a huge number of locations, scales and aspect ratios. Very slow! Need to do ~2k independent forward passes for each image!



Fast R-CNN



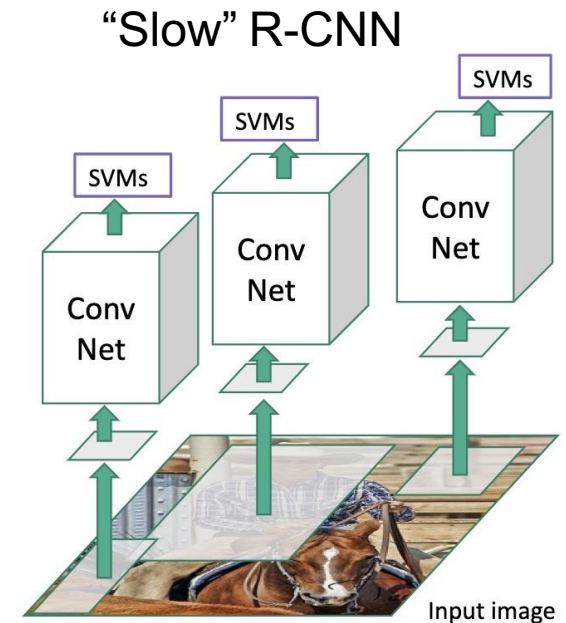
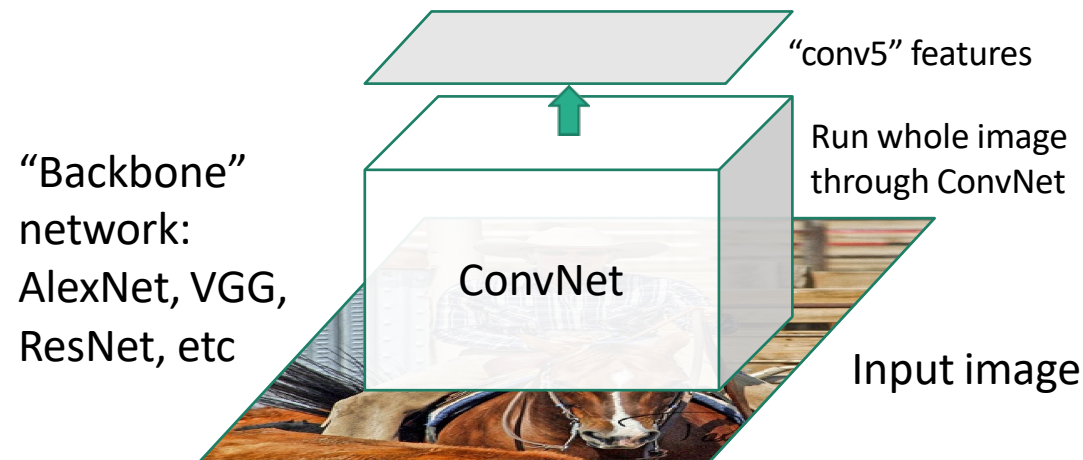
Input image



Idea: Pass the image through the model before cropping!



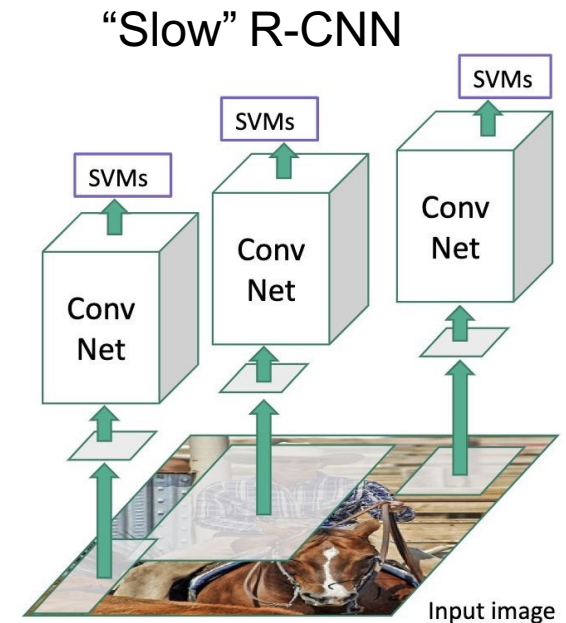
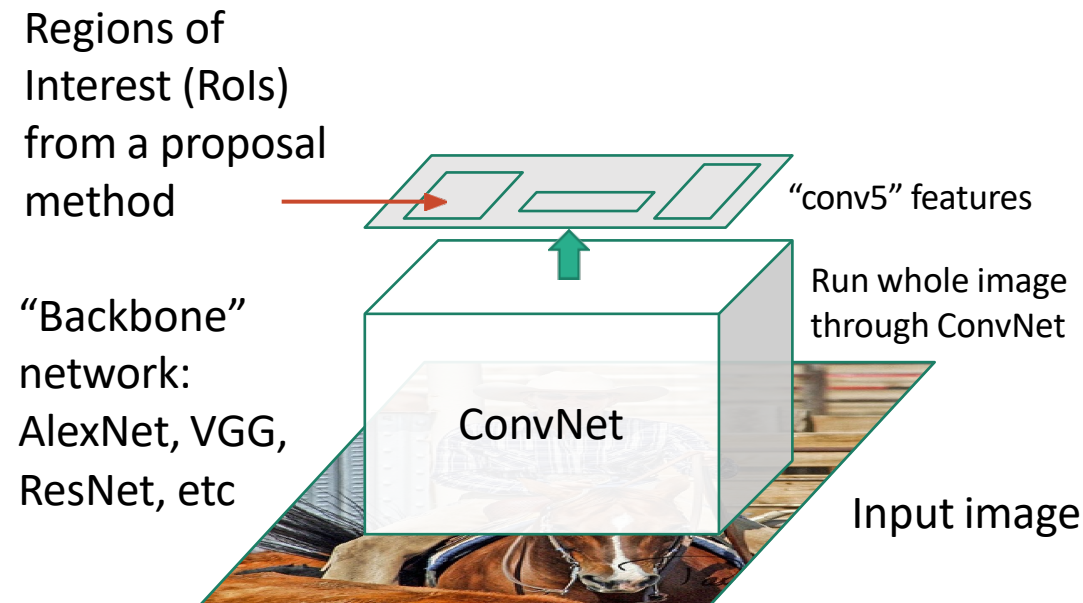
Fast R-CNN



Idea: Pass the image through the model before cropping!



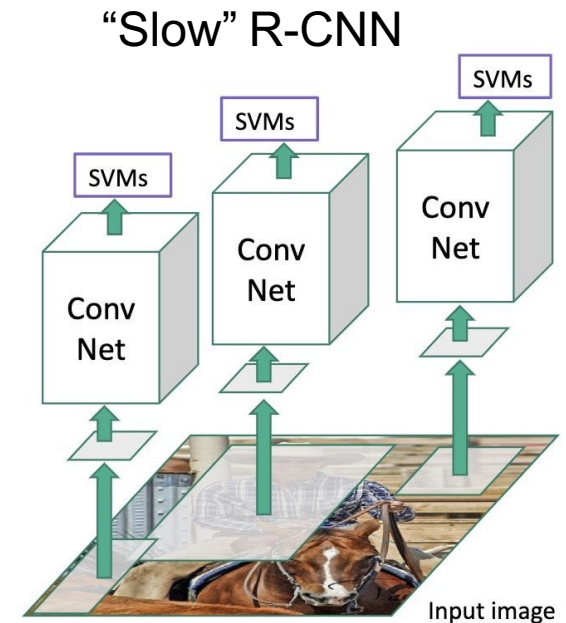
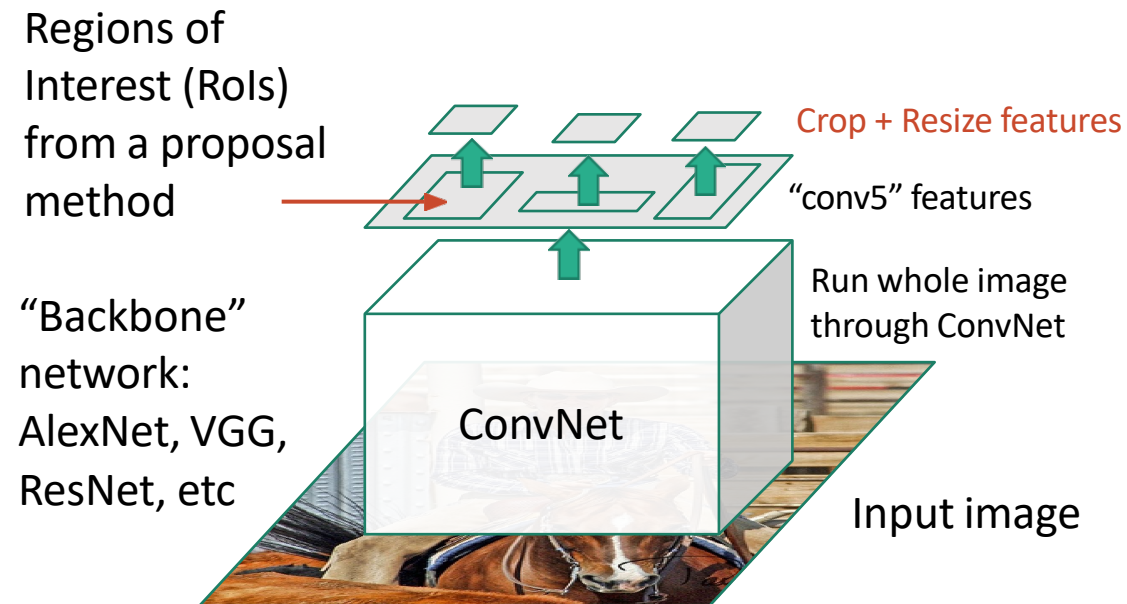
Fast R-CNN



Idea: Pass the image through the model before cropping!



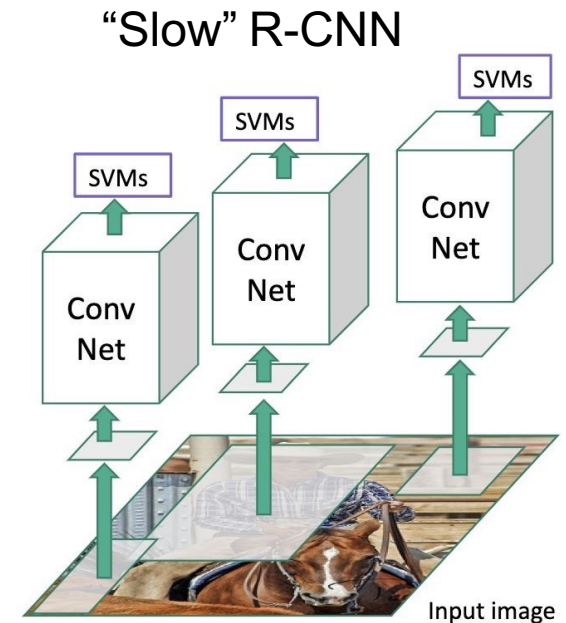
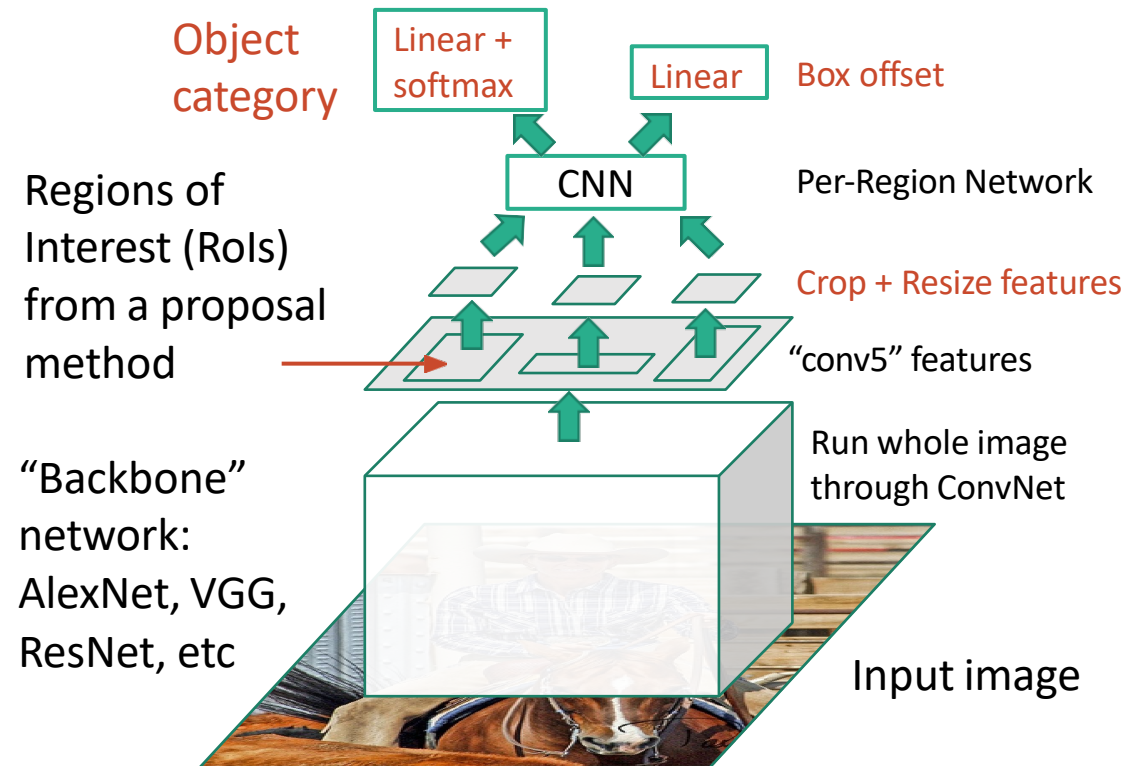
Fast R-CNN



Idea: Pass the image through the model before cropping!



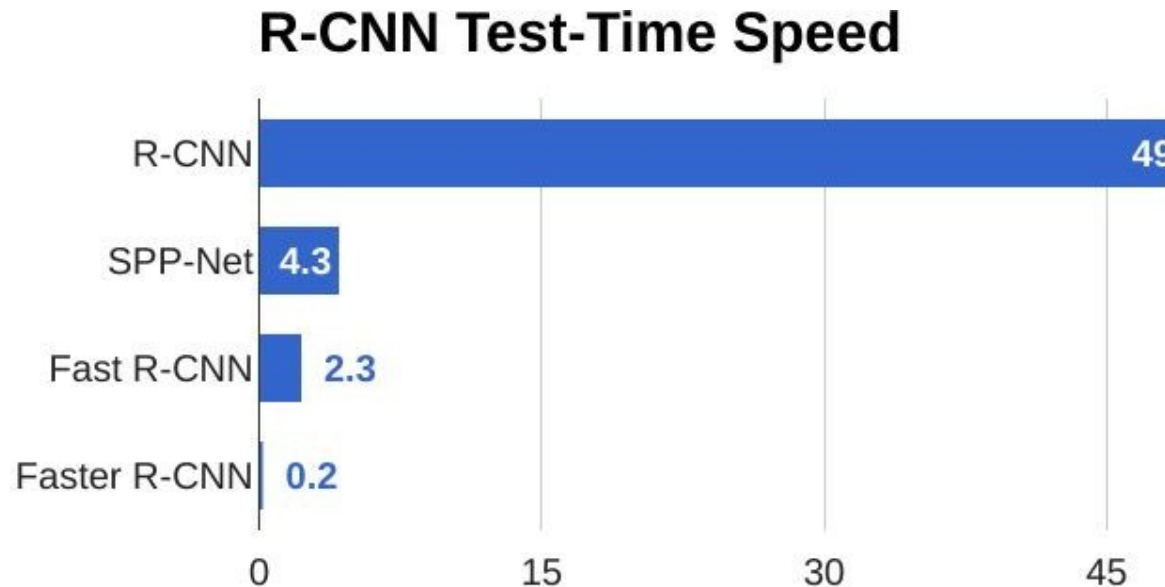
Fast R-CNN



Idea: Pass the image through the model before cropping!



Detection speed



Faster R-CNN:

- Learn a more accurate region proposal network. (Predictions at 5 FPS)

For real-time object detection, single stage detectors are often used (YOLO, SSD, RetinaNet)

- Faster but less accurate

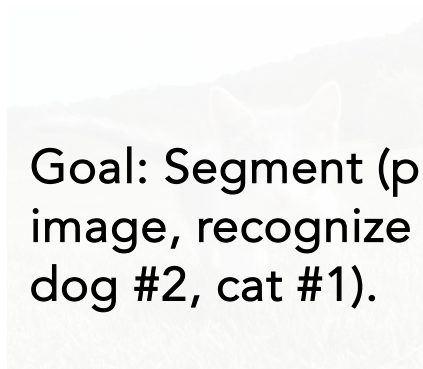
Redmon et al, "You Only Look Once: Unified, Real-Time Object Detection", CVPR 2016

Liu et al, "SSD: Single-Shot MultiBox Detector", ECCV 2016

Lin et al, "Focal Loss for Dense Object Detection", ICCV 2017

Computer Vision Tasks

Semantic Segmentation

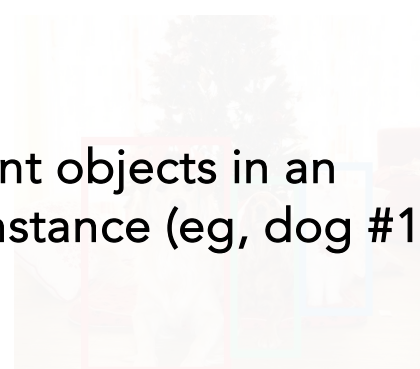


Goal: Segment (pixel-by-pixel) prominent objects in an image, recognize both their class and instance (eg, dog #1, dog #2, cat #1).

GRASS, CAT,
TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation

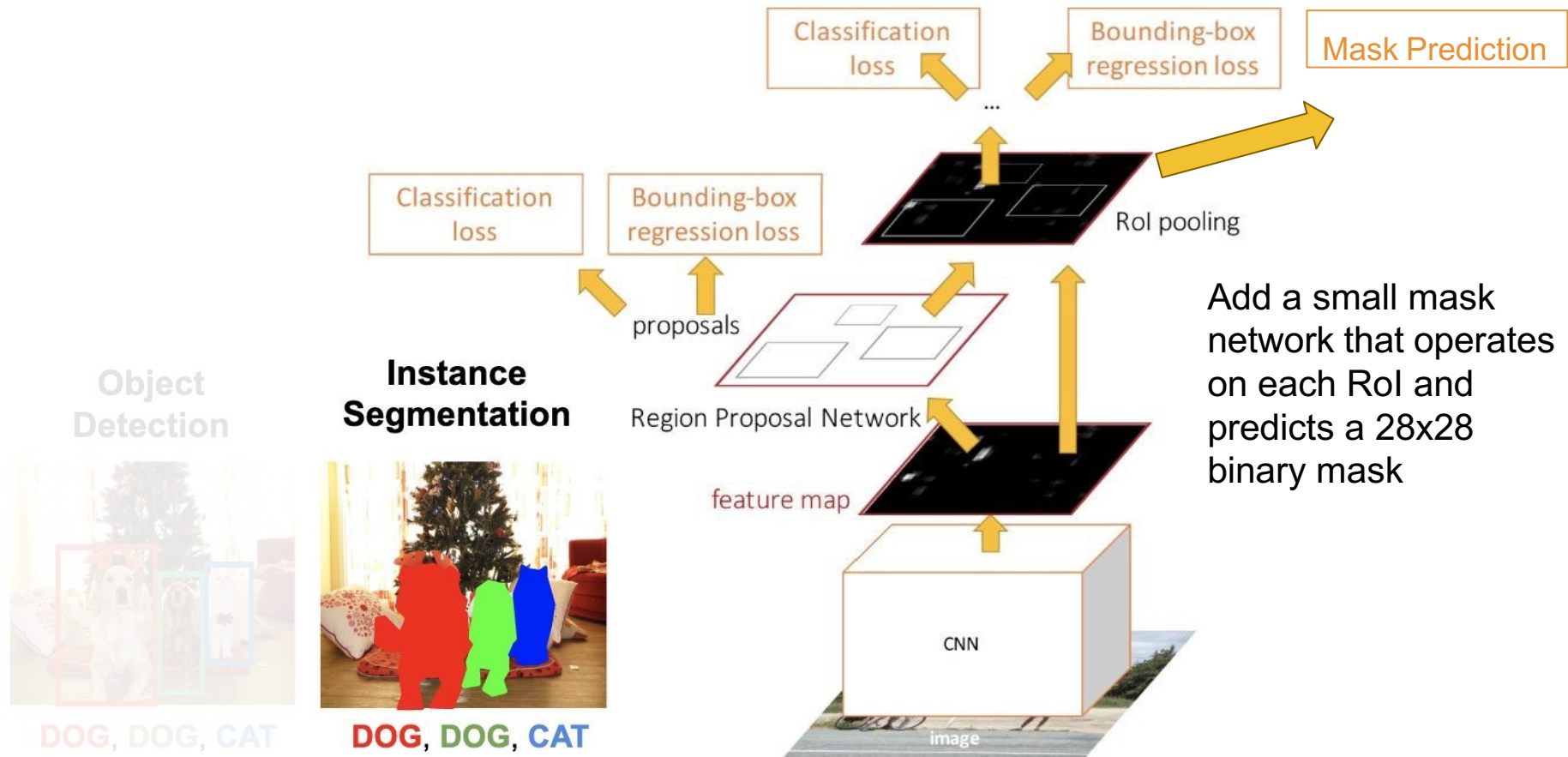


DOG, DOG, CAT

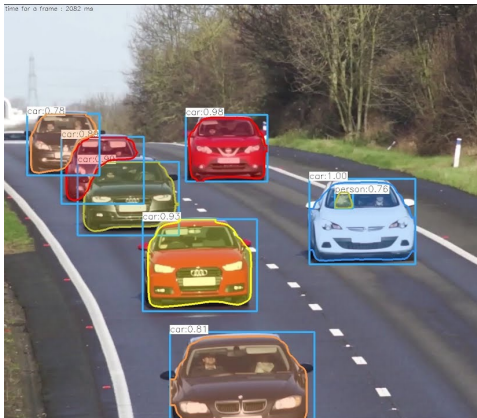
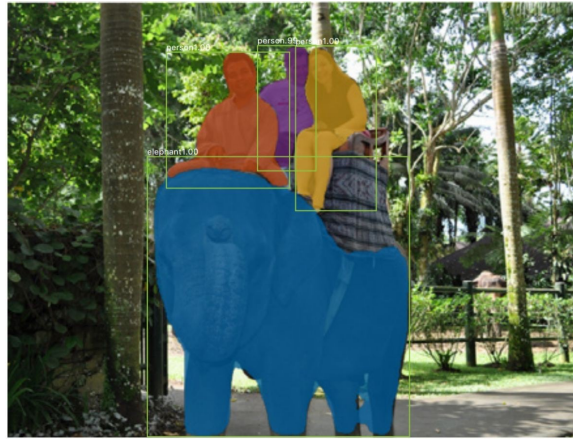
[This image is CC0 public domain](#)



Instance Segmentation: Mask R-CNN



Mask R-CNN: Outstanding Results!



Mask R-CNN also does pose detection



Open Source Frameworks

Lots of good implementations on GitHub!

TensorFlow Detection API:

https://github.com/tensorflow/models/tree/master/research/object_detection

Faster RCNN, SSD, RFCN, Mask R-CNN, ...

Detectron2 (PyTorch)

<https://github.com/facebookresearch/detectron2>

Mask R-CNN, RetinaNet, Faster R-CNN, RPN, Fast R-CNN, R-FCN, ...

Finetune on your own dataset with pre-trained models

